SEFI: A Reproducible Unified Geometric Framework for Cross-Domain Modeling

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August 20, 2026

Abstract

SEFI (Structured Emergent Field Integration) is a unified geometric framework designed for reproducible modeling across physical, biological, and cognitive domains. This interdisciplinary version emphasizes methodological transparency, computational implementability, and cross-domain applicability. We present the canonical SEFI warp-metric, operator set, SEFI:DNA biological model, and unified manifold evolution equation, along with reproducible examples and parameter choices. The goal is to provide a clear, testable foundation for researchers in multiple fields.

1 Introduction

Interdisciplinary research requires frameworks that are both conceptually coherent and computationally reproducible. SEFI provides such a framework by modeling systems as manifolds equipped with a warp-modified metric and a small set of operators governing field evolution. This paper presents SEFI in a form optimized for reproducibility and cross-domain use.

2 Methods

2.1 Geometric representation

Systems are represented as manifolds $\mathcal{M}$ with reference metric $g_{\mu\nu}$ and warp field $W(x)$. The canonical warp-metric is

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + \alpha W g_{\mu\nu} + \beta \nabla_\mu \nabla_\nu W.$$

2.2 Operator definitions

We define four operators acting on fields $\Phi(x)$:

- Warp: $\hat{W}\Phi = W\Phi$

- Tension: $\hat{T}\Phi = \nabla_\mu(T^\mu\Phi)$
- Coherence: $\hat{C}\Phi = C\Phi$
- Correction: $\hat{R}\Phi = -\gamma(\Phi - \Phi_{\text{eq}}) + \kappa\Delta\Phi$

2.3 Biological model

The SEFI:DNA evolution equation is

$$\partial_t\Psi = \hat{R}\Psi + \hat{D}\Psi - \hat{A}\Psi,$$

with explicit forms for duplication and aging.

2.4 Cosmological model

The unified evolution equation is

$$\partial_t\tilde{g}_{\mu\nu} = -2\kappa_G\left(R_{\mu\nu} - \tfrac{1}{2}R\tilde{g}_{\mu\nu}\right) + S_{\mu\nu}[W, T, C].$$

3 Results

We present reproducible examples:

- Warped surfaces and curvature maps
- SEFI:DNA simulations with parameter sweeps
- Cosmological toy models with SEFI sources

4 Discussion

We discuss cross-domain applicability, reproducibility, and limitations.

5 Conclusion

SEFI provides a reproducible geometric framework suitable for interdisciplinary research across physics, biology, and cognitive science.

References